



Overview

This chapter captures trends in AI R&D. It begins by examining AI publications, including journal articles, conference papers, and repositories. Next it considers data on significant machine learning systems, including large language and multimodal models. Finally, the chapter concludes by looking at AI conference attendance and open-source AI research. Although the United States and China continue to dominate AI R&D, research efforts are becoming increasingly geographically dispersed.



Chapter Highlights

The United States and China had the greatest number of cross-country collaborations in AI publications from 2010 to 2021, although the pace of collaboration has since slowed.

The number of AI research collaborations between the United States and China increased roughly 4 times since 2010, and was 2.5 times greater than the collaboration totals of the next nearest country pair, the United Kingdom and China. However, the total number of U.S.-China collaborations only increased by 2.1% from 2020 to 2021, the smallest year-over-year growth rate since 2010.

Industry races ahead of academia.

Until 2014, most significant machine learning models were released by academia. Since then, industry has taken over. In 2022, there were 32 significant industry-produced machine learning models compared to just three produced by academia. Building state-of-the-art AI systems increasingly requires large amounts of data, computer power, and money—resources that industry actors inherently possess in greater amounts compared to nonprofits and academia.

AI research is on the rise, across the board. The total number of AI publications has more than doubled since 2010. The specific AI topics that continue to dominate research include pattern recognition, machine learning, and computer vision.

China continues to lead in total AI journal, conference, and repository publications.

The United States is still ahead in terms of AI conference and repository citations, but those leads are slowly eroding. Still, the majority of the world's large language and multimodal models (54% in 2022) are produced by American institutions.

Large language models are getting bigger and more expensive.

GPT-2, released in 2019, considered by many to be the first large language model, had 1.5 billion parameters and cost an estimated \$50,000 USD to train. PaLM, one of the flagship large language models launched in 2022, had 540 billion parameters and cost an estimated \$8 million USD—PaLM was around 360 times larger than GPT-2 and cost 160 times more. It's not just PaLM: Across the board, large language and multimodal models are becoming larger and pricier.



This section draws on data from the Center for Security and Emerging Technology (CSET) at Georgetown University. CSET maintains a merged corpus of scholarly literature that includes Digital Science's Dimensions, Clarivate's Web of Science, Microsoft Academic Graph, China National Knowledge Infrastructure, arXiv, and Papers With Code. In that corpus, CSET applied a classifier to identify English-language publications related to the development or application of AI and ML since 2010. For this year's report, CSET also used select Chinese AI keywords to identify Chinese-language AI papers; CSET did not deploy this method for previous iterations of the AI Index report.¹

In last year's edition of the report, publication trends were reported up to the year 2021. However, given that there is a significant lag in the collection of publication metadata, and that in some cases it takes until the middle of any given year to fully capture the previous year's publications, in this year's report, the AI Index team elected to examine publication trends only through 2021, which we, along with CSET, are confident yields a more fully representative report.

1.1 Publications

Overview

The figures below capture the total number of English-language and Chinese-language AI publications globally from 2010 to 2021—by type, affiliation, cross-country collaboration, and cross-industry collaboration. The section also breaks down

publication and citation data by region for AI journal articles, conference papers, repositories, and patents.

Total Number of AI Publications

Figure 1.1.1 shows the number of AI publications in the world. From 2010 to 2021, the total number of AI publications more than doubled, growing from 200,000 in 2010 to almost 500,000 in 2021.

Number of AI Publications in the World, 2010–21

Source: Center for Security and Emerging Technology, 2022 | Chart: 2023 AI Index Report

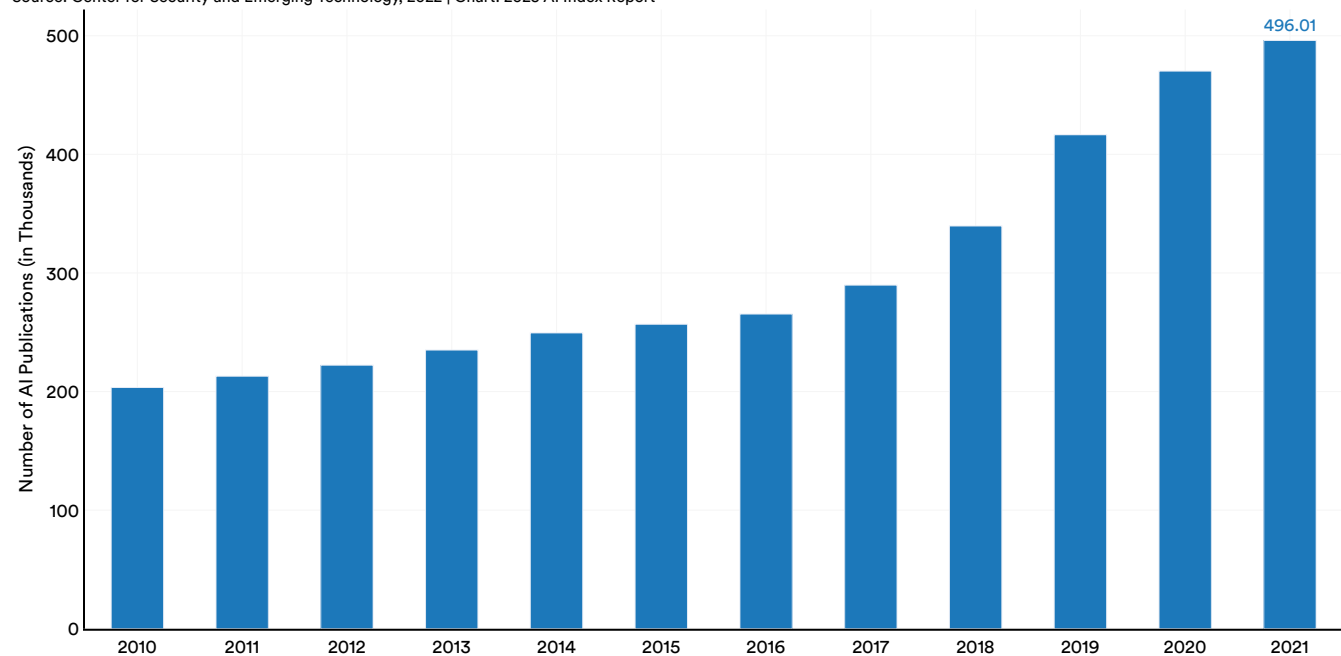


Figure 1.1.1

¹ See the Appendix for more information on CSET's methodology. For more on the challenge of defining AI and correctly capturing relevant bibliometric data, see the AI Index team's discussion in the paper "[Measurement in AI Policy: Opportunities and Challenges](#)."



By Type of Publication

Figure 1.1.2 shows the types of AI publications released globally over time. In 2021, 60% of all published AI documents were journal articles, 17% were conference papers, and 13% were repository submissions. Books,

book chapters, theses, and unknown document types made up the remaining 10% of publications. While journal and repository publications have grown 3 and 26.6 times, respectively, in the past 12 years, the number of conference papers has declined since 2019.

Number of AI Publications by Type, 2010–21

Source: Center for Security and Emerging Technology, 2022 | Chart: 2023 AI Index Report

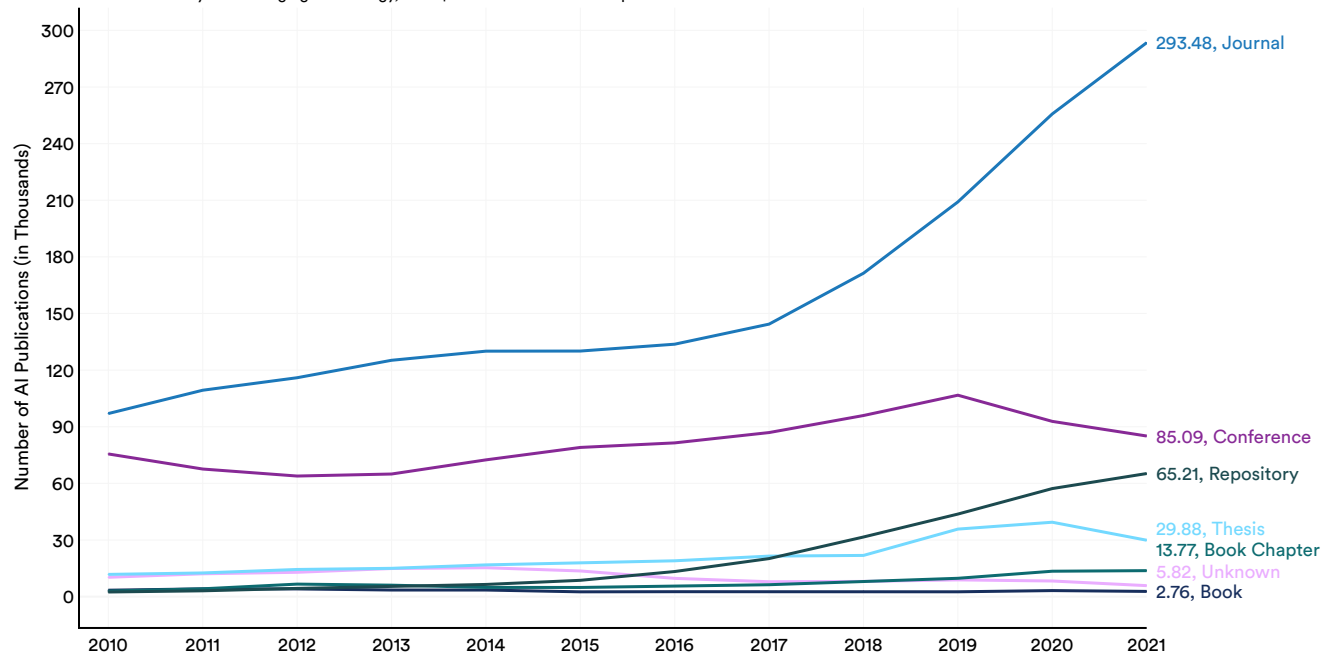


Figure 1.1.2

Narrative Highlight:**Top Publishing Institutions****All Fields**

Since 2010, the institution producing the greatest number of total AI papers has been the Chinese Academy of Sciences (Figure 1.1.21). The next top four are all Chinese universities: Tsinghua

University, the University of the Chinese Academy of Sciences, Shanghai Jiao Tong University, and Zhejiang University.⁵ The total number of publications released by each of these institutions in 2021 is displayed in Figure 1.1.22.

Top Ten Institutions in the World in 2021 Ranked by Number of AI Publications in All Fields, 2010–21

Source: Center for Security and Emerging Technology, 2022 | Chart: 2023 AI Index Report

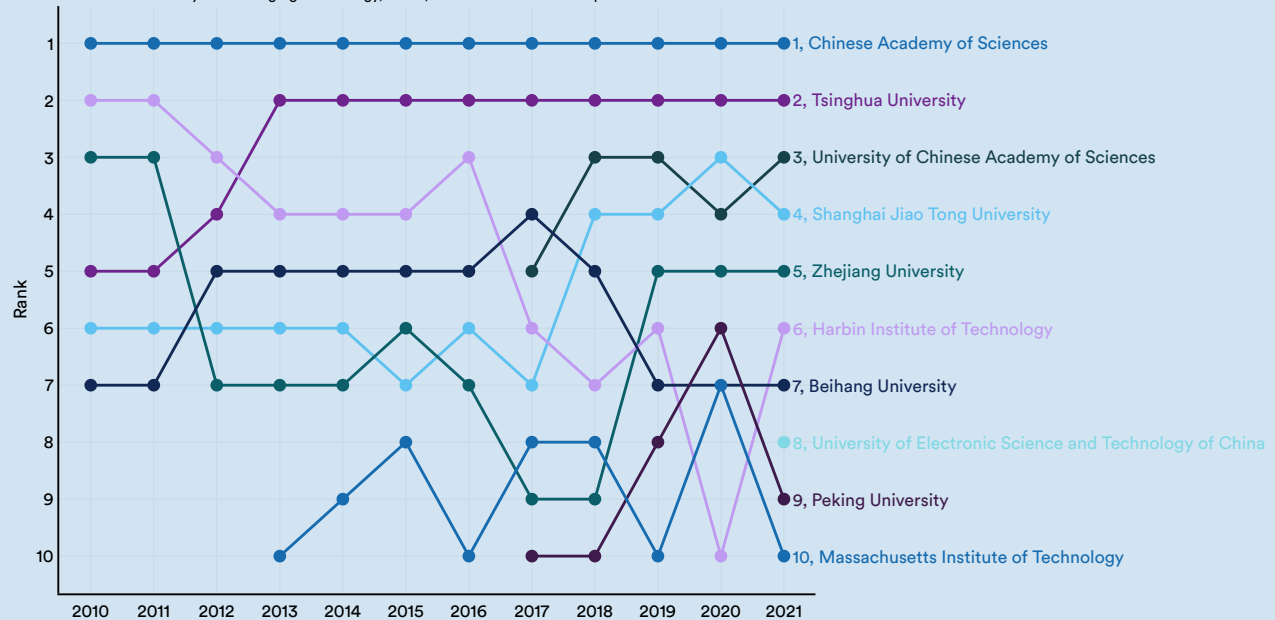


Figure 1.1.21

⁵ It is important to note that many Chinese research institutions are large, centralized organizations with thousands of researchers. It is therefore not entirely surprising that, purely by the metric of publication count, they outpublish most non-Chinese institutions.



Narrative Highlight:

Top Publishing Institutions (cont'd)

Top Ten Institutions in the World by Number of AI Publications in All Fields, 2021

Source: Center for Security and Emerging Technology, 2022 | Chart: 2023 AI Index Report

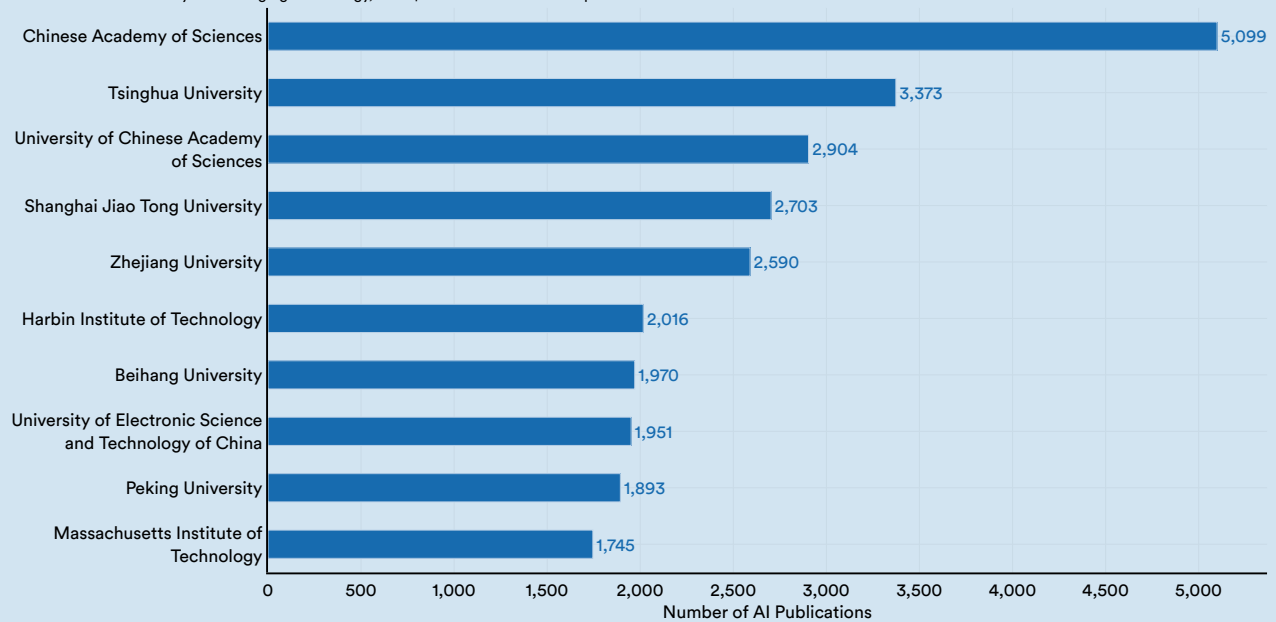


Figure 1.1.22



Epoch AI is a collective of researchers investigating and forecasting the development of advanced AI. Epoch curates a database of significant AI and machine learning systems that have been released since the 1950s. There are different criteria under which the Epoch team decides to include particular AI systems in their database; for example, the system may have registered a state-of-the-art improvement, been deemed to have been historically significant, or been highly cited.

This subsection uses the Epoch database to track trends in significant AI and machine learning systems. The latter half of the chapter includes research done by the AI Index team that reports trends in large language and multimodal models, which are models trained on large amounts of data and adaptable to a variety of downstream applications.

1.2 Trends in Significant Machine Learning Systems

General Machine Learning Systems

The figures below report trends among all machine learning systems included in the Epoch dataset. For reference, these systems are referred to as *significant machine learning systems* throughout the subsection.

System Types

Among the significant AI machine learning systems released in 2022, the most common class of system was language (Figure 1.2.1). There were 23 significant AI language systems released in 2022, roughly six times the number of the next most common system type, multimodal systems.

Number of Significant Machine Learning Systems by Domain, 2022

Source: Epoch, 2022 | Chart: 2023 AI Index Report

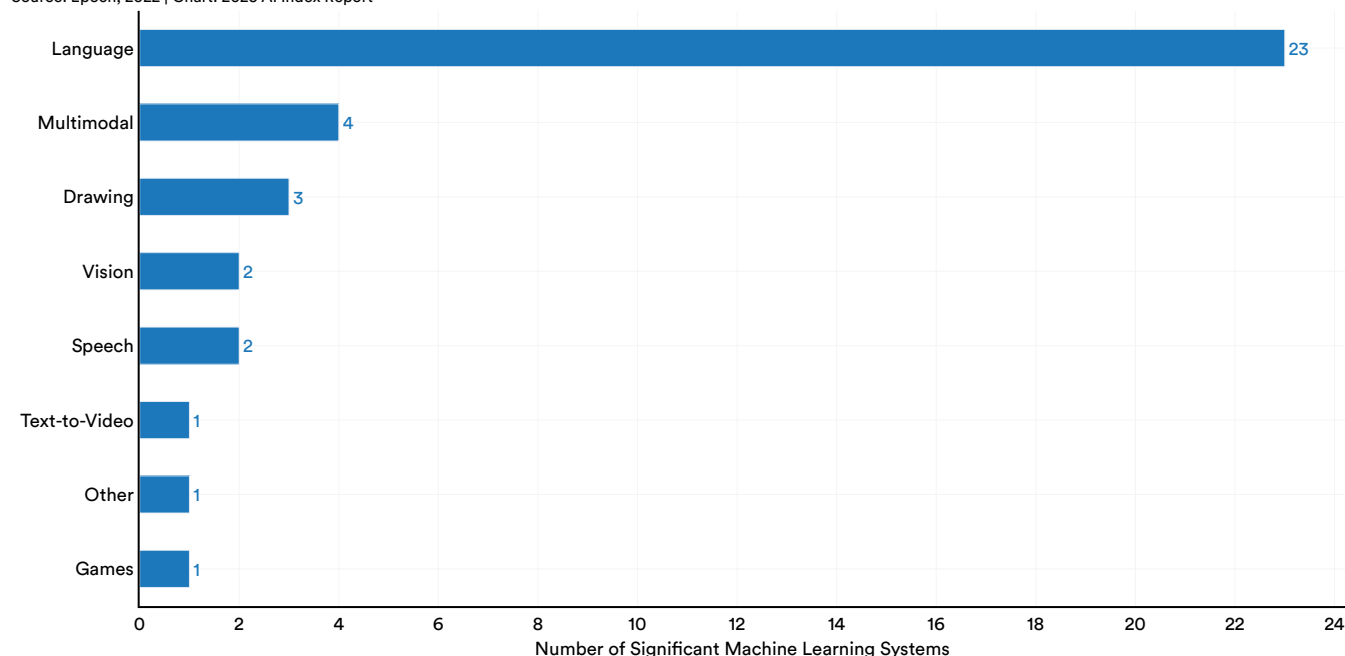


Figure 1.2.1⁶

⁶ There were 38 total significant AI machine learning systems released in 2022, according to Epoch; however, one of the systems, BaGuaLu, did not have a domain classification and is therefore omitted from Figure 1.2.1.

National Affiliation

In order to paint a picture of AI’s evolving geopolitical landscape, the AI Index research team identified the nationality of the authors who contributed to the development of each significant machine learning system in the Epoch dataset.⁷

Systems

Figure 1.2.3 showcases the total number of significant machine learning systems attributed to researchers from particular countries.⁸ A researcher is considered to have belonged to the country in which their institution, for example a university

or AI-research firm, was headquartered. In 2022, the United States produced the greatest number of significant machine learning systems with 16, followed by the United Kingdom (8) and China (3). Moreover, since 2002 the United States has outpaced the United Kingdom and the European Union, as well as China, in terms of the total number of significant machine learning systems produced (Figure 1.2.4). Figure 1.2.5 displays the total number of significant machine learning systems produced by country since 2002 for the entire world.

Number of Significant Machine Learning Systems by Country, 2022

Source: Epoch and AI Index, 2022 | Chart: 2023 AI Index Report

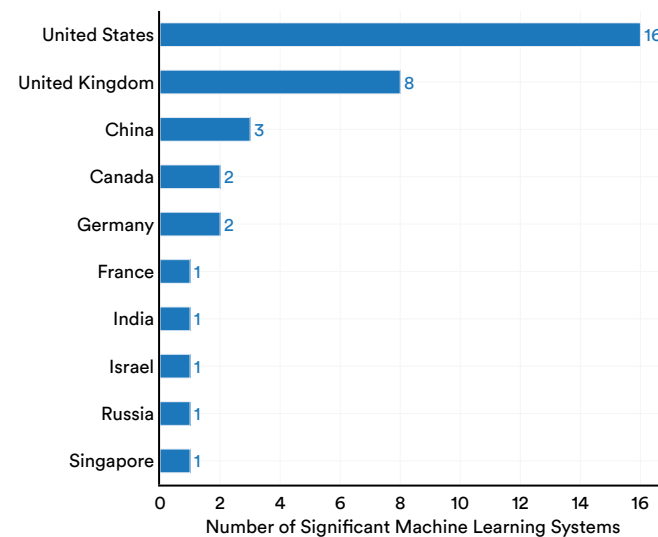


Figure 1.2.3

Number of Significant Machine Learning Systems by Select Geographic Area, 2002–22

Source: Epoch and AI Index, 2022 | Chart: 2023 AI Index Report

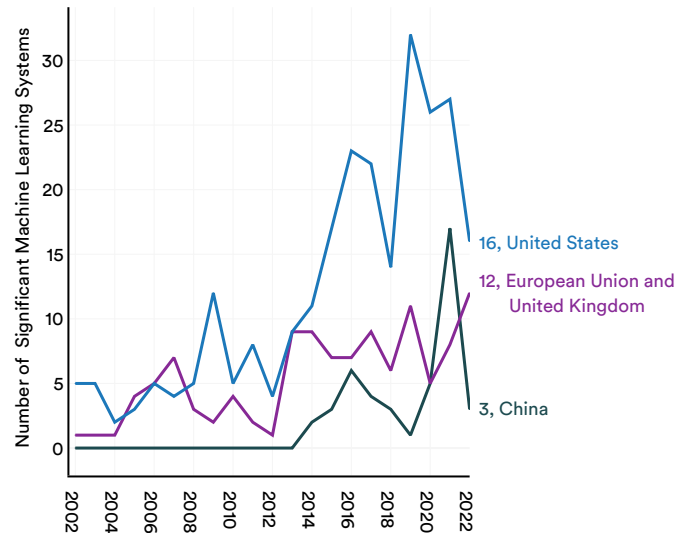


Figure 1.2.4

⁷ The methodology by which the AI Index identified authors’ nationality is outlined in greater detail in the Appendix.

⁸ A machine learning system is considered to be affiliated with a particular country if at least one author involved in creating the model was affiliated with that country. Consequently, in cases where a system has authors from multiple countries, double counting may occur.